strategy of estimating our own price. This is because the two houses are similar in attributes and physically close to each other, so we expect them to sell for similar amounts. Alix's house, on the other hand, is in California and is entirely irrelevant to our house, so even if we know for how much she sold her house, we won't be able to gain any new information about what we're interested in: how much our own house is worth.

The calculation we just went through is a Bayesian update to our belief about the price of our house. You might be familiar with Bayes' theorem, which comes in the form in Figure 2.2. For an excellent introduction to the Bayes' theorem and Bayesian learning, check out Chapter 8 of Luis Serrano's *Grokking Machine Learning*.

Bayes' theorem gives us a way of updating our belief about a quantity that we're interested in, which in this case is the appropriate price for our house. When applying Bayes' theorem, we go from our prior belief, which is our first guess, to a posterior belief about the quantity in question. This posterior belief combines the prior and the likelihood of any data we observe.

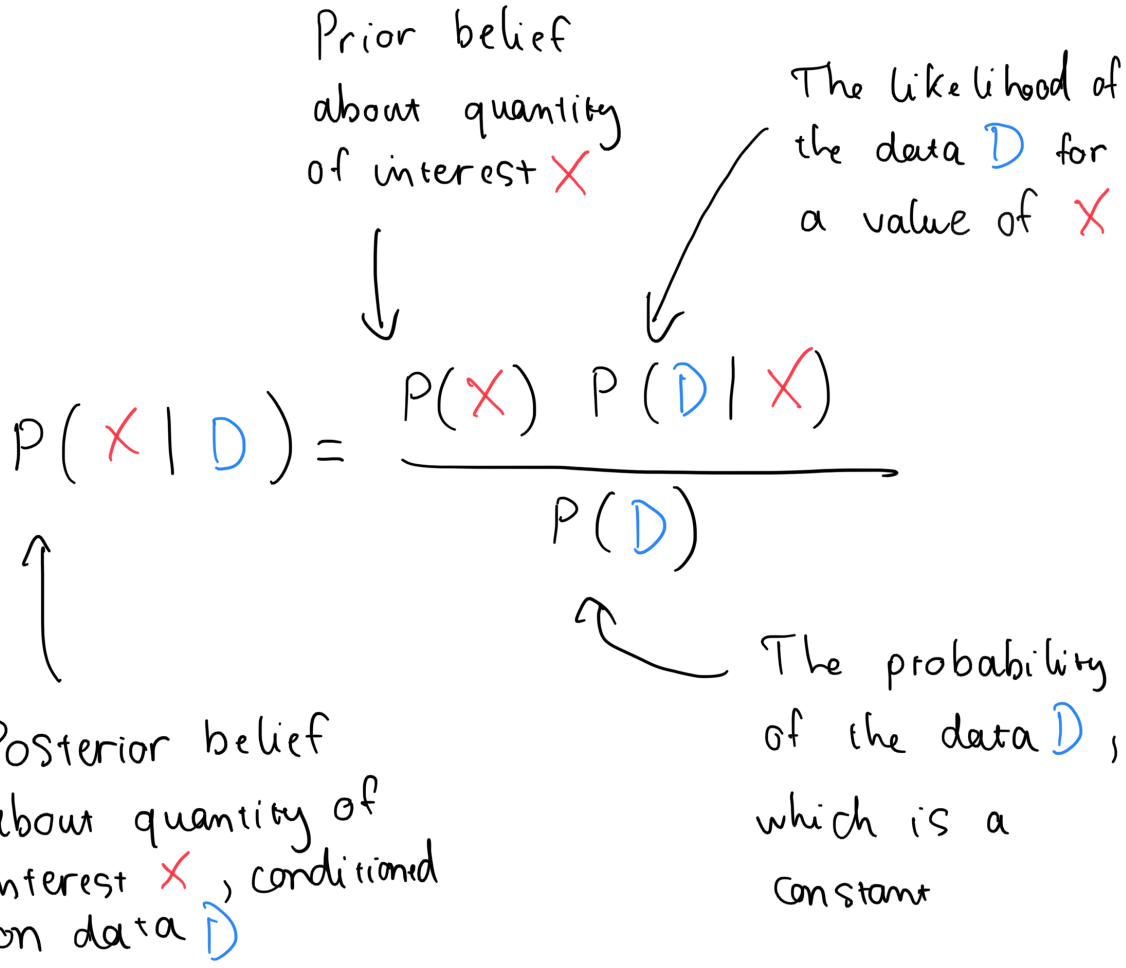


Figure 2.2 Bayes' theorem, which gives a method of updating a belief about a quantity of interest, represented as a probability distribution of a random variable. Before observing any data, we have the prior belief about X. After being updated with data, we obtain the posterior belief about X.

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